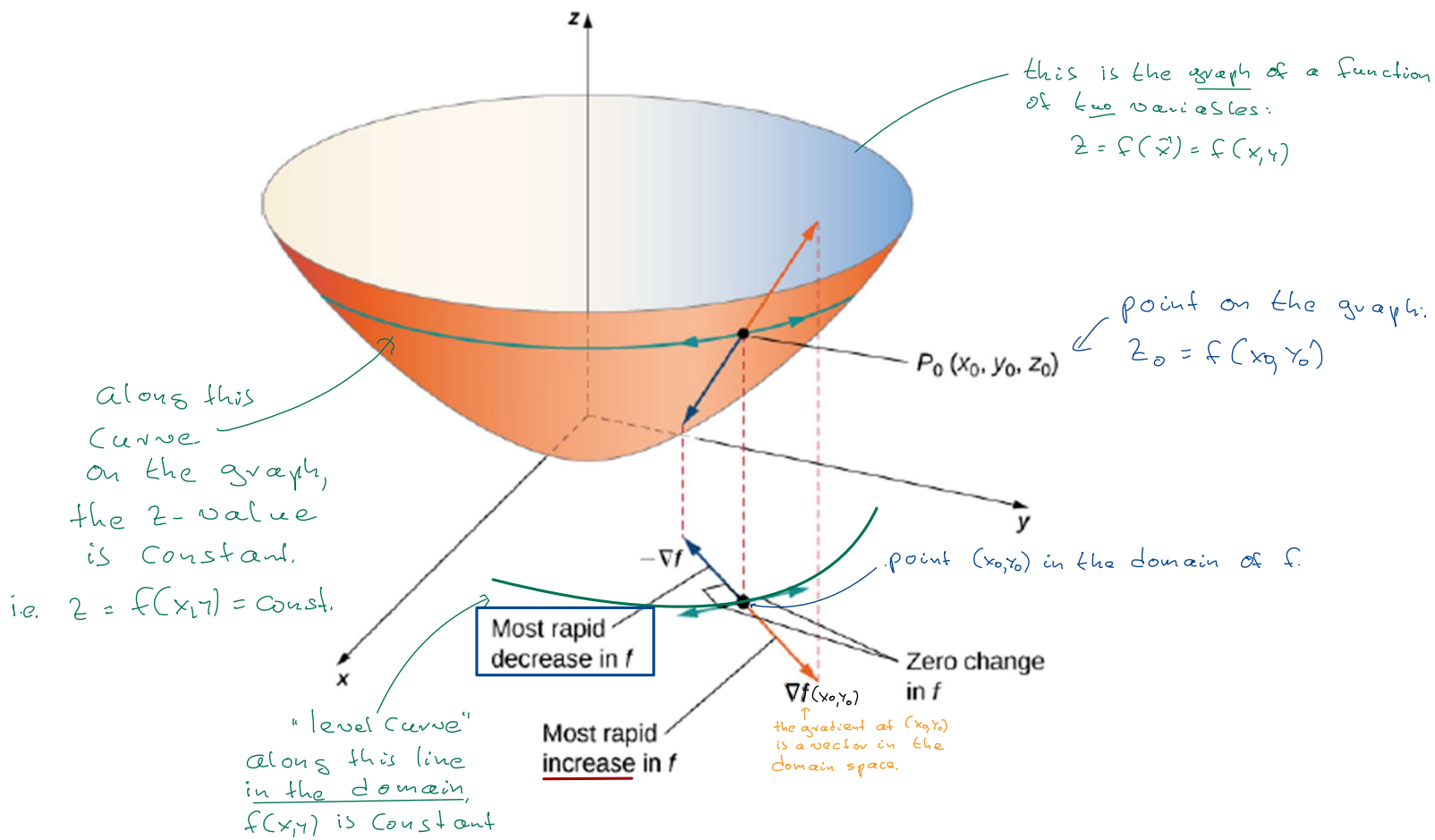
IV.6 The Steepest Descent Method

To simplify discussion, we only will discuss the problem of finding a **minimum** of a function $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ of several variables.



Recall from Calculus 2: The gradient $\nabla f(\mathbf{x}_0)$ at a point $\mathbf{x}_0$ points in direction of the *steepest increase* of the function f.

[most rapid]

Thus, the negative of the gradient, $-\nabla f(\mathbf{x}_0)$, points in direction of the *steepest descent* of f. So if we want to find the minimizer of $f(\mathbf{x})$, then we should move away from $\mathbf{x}_0$, in direction of $-\nabla f(\mathbf{x}_0)$. This gives the following iterations *(formula for the steepest decent method)*:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$$

from $\vec{x_k}$, move in direction of $-\nabla f(\vec{x_k})$

where $\alpha > 0$. Question: how large should α be ?

Observe: As α increases, the point $\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$ moves along a straight line in the domain of f.

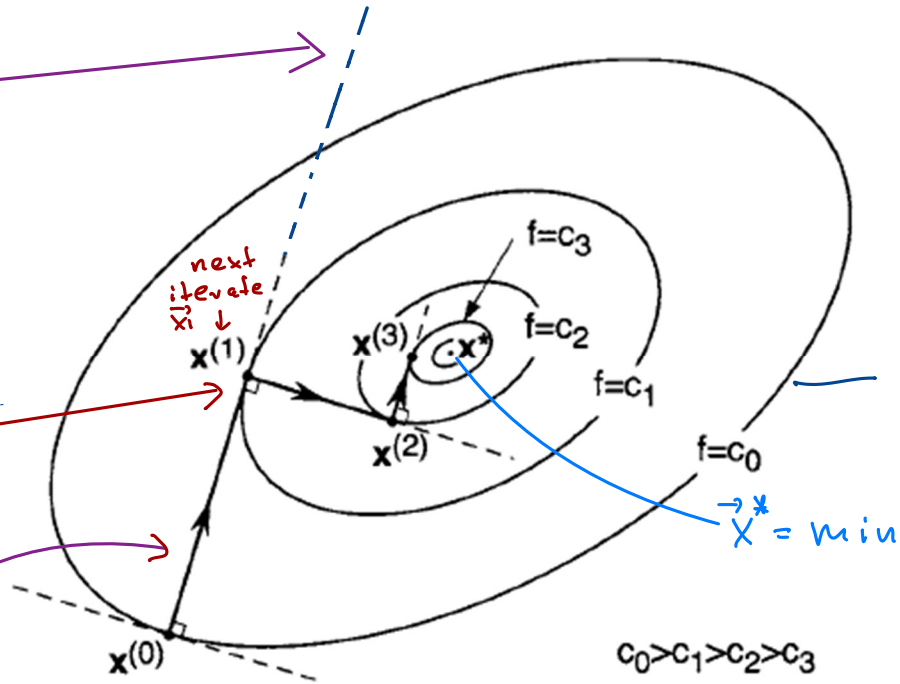
At the same time, the point $\left(\mathbf{x}_k,\ f\left(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)\right)\right)$ moves along a curve on the graph of f in decreasing direction. We should stop when this point is lowest, that is, where $f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k))$ is smallest.

Sketch in 2 dimensions: [Domain of f]

line with vector equation

$\vec{x} = \vec{x}_0 - \alpha \nabla f(\vec{x}_0)$

$g(\alpha) = f(\vec{x}_0 - \alpha \nabla f(\vec{x}_0))$ is smallest here



level curve $f(\vec{x}) = f(x,y) = c_0$

$\vec{x}^* =$ minimizer

direction of $-\nabla f(\vec{x}_0)$

initial point $\vec{x}_0$

tangent line to the level curve at $\vec{x}_0$

That is, the number α should be a minimizer of the function of *one variable*

$$g_k(\alpha) = f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)), \qquad \alpha > 0$$

(One can find the value of α (at each iteration k) by using Newton's method, or the secant method, as descriped in section IV.5)

Thus, the ***steepest descent algorithm*** is as follows:

1. First find

α_k is the minimizer of $g_k(\alpha)$

$$\alpha_k = \arg\min_{\alpha} \underbrace{f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k))}_{g_k(\alpha)}, \qquad \alpha > 0$$

$g_k(\alpha)$, a function of <u>one</u> variable α. It is easy to find its minimizer α_k.

2. Then set

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$$

Stopping criteria:

1. Because $\nabla f(\mathbf{x}) = 0$ at a minimizer, one stops when the gradient is a small, i.e. when $\|\nabla f(\mathbf{x}_k)\| < \epsilon$, where ϵ is a prespecified threshold.
2. One may also stop when the iterations don't change much any more, i.e. when the *relative change* is small, $\dfrac{\|\mathbf{x}_{k+1} - \mathbf{x}_k\|}{\|\mathbf{x}_k\|} < \epsilon$. [Sometimes $\|x_{k+1} - x_k\| < \varepsilon$]

   "Relative distance"
3. Or one stops when the values of the function values don't change much any more, i.e. when $\dfrac{|f(\mathbf{x}_{k+1}) - f(\mathbf{x}_k)|}{|f(\mathbf{x}_k)|} < \epsilon$.

   [Sometimes $|f(x_{k+1}) - f(x_k)| < \varepsilon$]

Example 13: Use the steepest descent method to find the minimizer of

(*) $f(x, y, z) = (x-4)^4 + (y-3)^2 + 4(z+5)^4$ starting with $\mathbf{x}_0 = \langle 4, 2, -1 \rangle$.

Sol. (To save space, we write vectors in row form as in Calculus 2.)

Clearly,

(1) $$\nabla f(\mathbf{x}) = \nabla f(x, y, z) = 4(x-4)^3 + 2(y-3) + 16(z+5)^3$$

and we also need

$\vec{x} = (x, y, z)$

(2) $$g(\alpha) = f\big(\mathbf{x} - \alpha \nabla f(\mathbf{x})\big) = f\big((x, y, z) - \alpha \nabla f(x, y, z)\big)$$

(we don't compute further, but let the computer do)

Note: To make notations easy, we don't distinguish between vector and point.
(x, y, z) and $\langle x, y, z \rangle$ have the same meaning.

$k=0$: First compute α_0: With $\mathbf{x}_0 = \langle \overset{x_0}{4}, \overset{y_0}{2}, \overset{z_0}{-1} \rangle$ we get by (1) and (2):

$$\nabla f(\mathbf{x}_0) = \nabla f(x_0, y_0, z_0) \overset{(1)}{=} \left\langle 4(\underbrace{\overset{x_0=4}{4}-4}_{0})^3,\ 2(\underbrace{\overset{y_0=2}{2}-3}_{(-1)}),\ 16(\underbrace{\overset{z_0=-1}{-1}+5}_{4})^3 \right\rangle = \langle 0, -2, 1024 \rangle$$

$$\mathbf{x}_0 - \alpha \nabla f(\mathbf{x}_0) = (x_0, y_0, z_0) - \alpha \nabla f(x_0, y_0, z_0) = (4, 2, -1) - \alpha(0, -2, 1024)$$

$$= (4,\ 2+2\alpha,\ -1-1024\alpha)$$

$$g_0(\alpha) = f(\mathbf{x}_0 - \alpha \nabla f(\mathbf{x}_0)) = f(\ 4,\ 2+2\alpha,\ -1-1024\alpha\ )$$

plug into f

$$\overset{(*)}{=} (\underbrace{4-4}_{=0})^4 + (\underbrace{[2+2\alpha]-3}_{2\alpha-1})^2 + 4(\underbrace{[-1-1024\alpha]+5}_{4-1024\alpha})^4$$

$$= (2\alpha-1)^2 + 4(\ 4-1024\alpha\ )^4$$

Using Newton's method or the secant method, one obtains that

$$g_0'(\alpha) = 0 \quad \text{when} \quad \alpha_0 = \alpha = 0.003967.$$

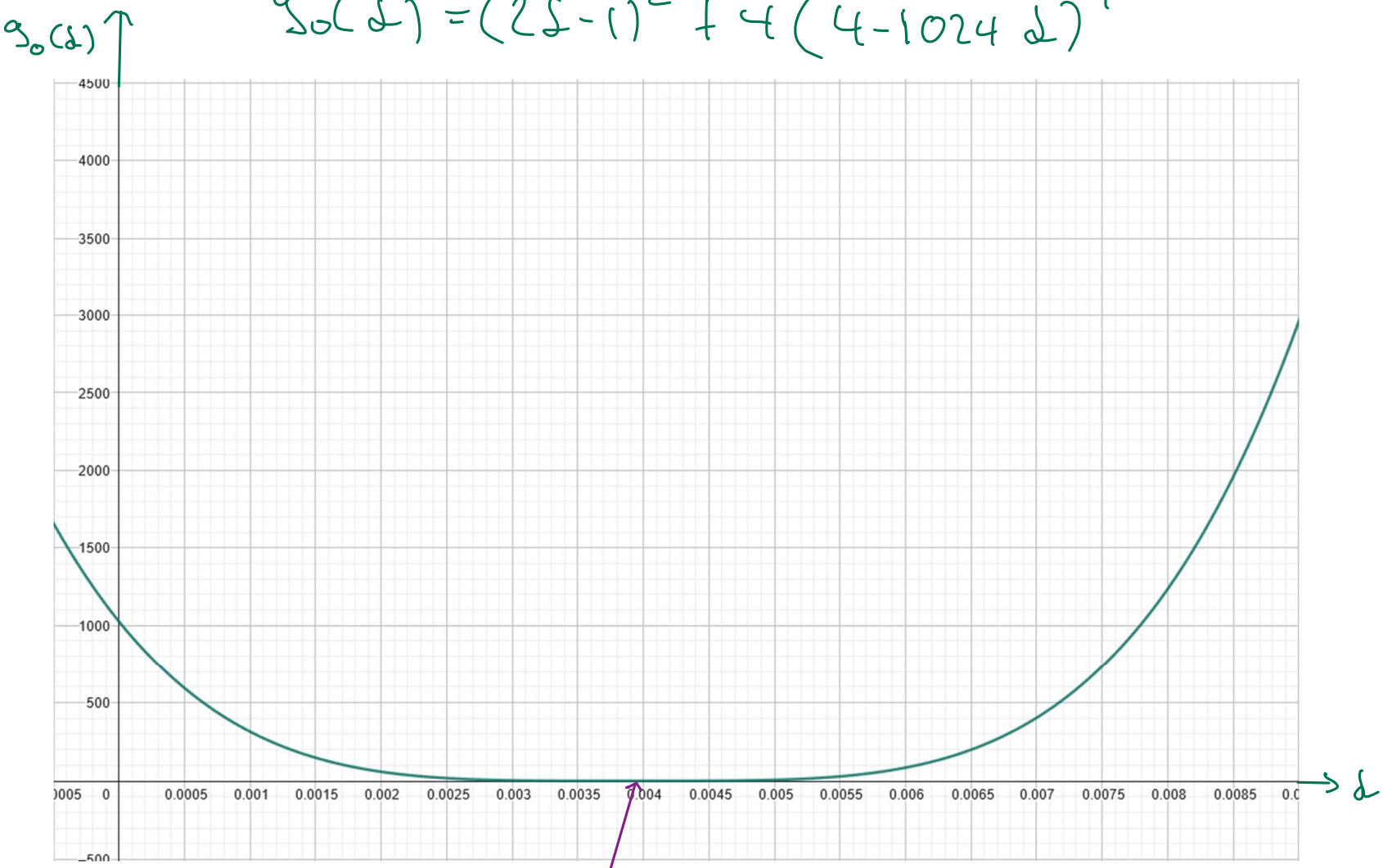
Use an iterative method for finding a root of g' [Not shown]

Then $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0) = \langle 4, 2, -1 \rangle - 0.003967 \langle 0, -2, 1024 \rangle$

$$= \langle 4.000,\ 2.008,\ -5.062 \rangle$$

This is the graph of

$$g_0(\delta) = (2\delta - 1)^2 + 4(4 - 1024\,\delta)^4$$



Min. at 0.003967

$\delta_0 = 0.003967$

$k=1$: First compute α_1: With $\mathbf{x}_1 = \langle 4,\ 2.008,\ -5.062 \rangle$ we get

$$\nabla f(\mathbf{x}_1) = \nabla f(x_1, y_1, z_1) \overset{(1)}{=} \left\langle 4(4-4)^3,\ 2(2.008-3),\ 16(-5.062+5)^3 \right\rangle$$

$$= \langle 0,\ -1.984,\ -0.003875 \rangle$$

$$\mathbf{x}_1 - \alpha \nabla f(\mathbf{x}_1) = (\overset{x_1}{4},\ \overset{y_1}{2.008},\ \overset{z_1}{-5.062}) - \alpha(0,\ -1.984,\ -0.003875)$$

$$= (\ 4,\ \ 2.008 + 1.984\alpha,\ \ -5.062 + 0.003875\alpha\ )$$

$$g_1(\alpha) = f\big(\ 4,\ \ 2.008 + 1.984\alpha,\ \ -5.062 + 0.003875\alpha\ \big)$$

$$\overset{(*)}{=} (4-4)^4 + \big(\ [2.008 + 1.984\alpha] - 3\big)^2 + 4\big(\ [-5.062 + 0.003875\alpha] + 5\ \big)^4$$

$$= \big(\ 1.984\alpha - 0.992\big)^2 + 4\big(\ 0.003875\alpha - 0.062\ \big)^4$$

Using Netwon's method or the secant method, we obtain that

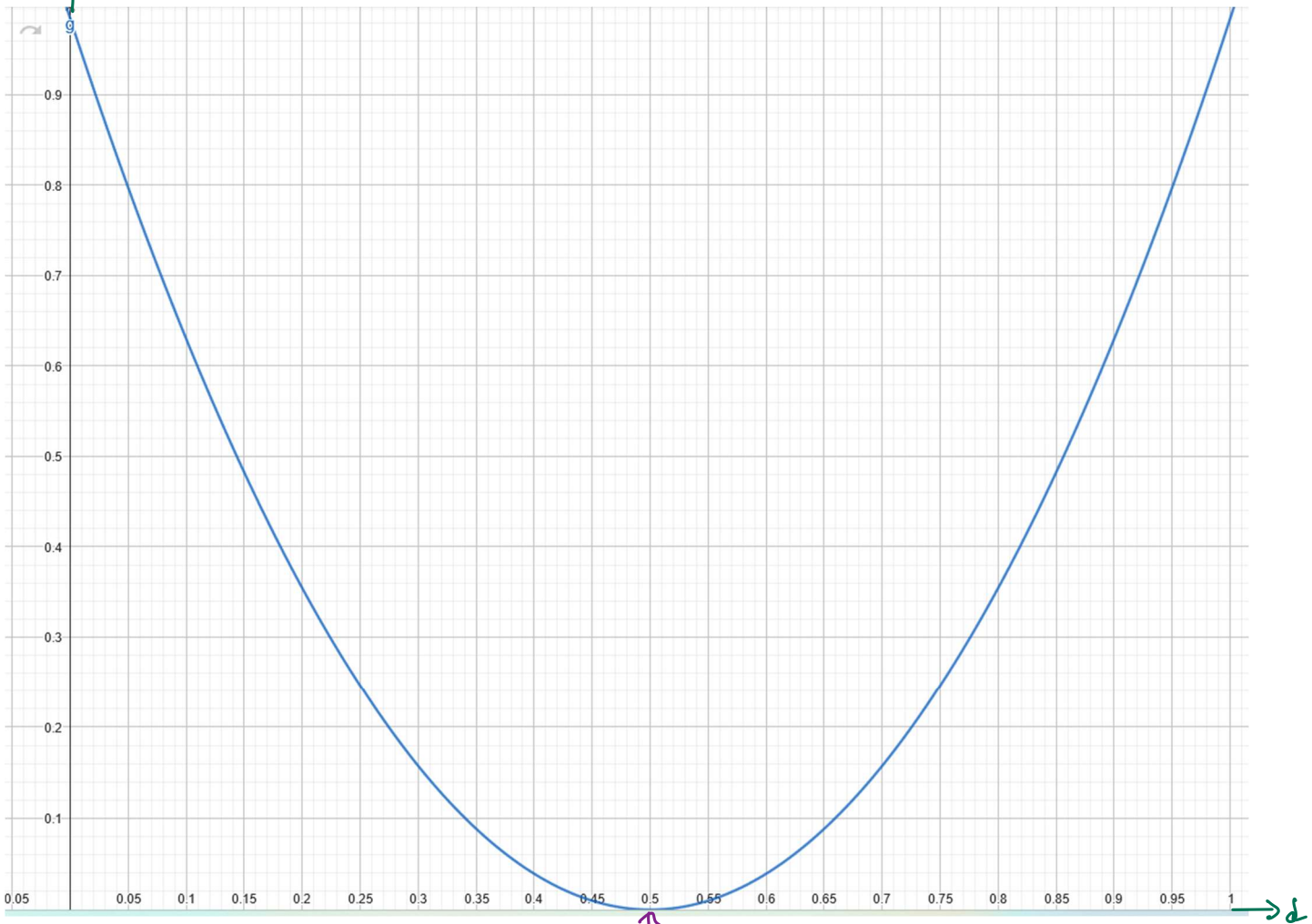
$$g_1'(\alpha) = 0 \quad \text{when} \quad \alpha_1 = \alpha = 0.5000.$$ (details not shown)

$$\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1\ \nabla f(\mathbf{x}_1) = \langle 4.0, 2.008, -5.062 \rangle - 0.50 \langle 0, -1.984,\ -0.003875 \rangle$$

$$= \langle 4.000,\ 3.000,\ -5.060 \rangle$$

$g_1(d)$

graph of $g_1(d) = (1.984d - 0.992)^2 + 4(0.003875d - 0.062)^4$



Min at ≈ 0.5

$d_1 = 0.5$

$k=2$: First compute α_1: With $\mathbf{x}_2 = \langle 4, 3, -5.060 \rangle$ we get

$$\nabla f(\mathbf{x}_2) = \nabla f(x_2, y_2, z_2) \overset{(1)}{=} \left\langle 4(4-4)^3,\ 2(3-3),\ 16(-5.06+5)^3 \right\rangle$$

$$= \langle 0, 0, -0.003525 \rangle$$

$$\mathbf{x}_2 - \alpha \nabla f(\mathbf{x}_2) = (4, 3, -5.060) - \alpha(0, 0, -0.003525)$$

$$= (4,\ 3,\ 0.003325\alpha - 5.06)$$

$$g_2(\alpha) = f\left(4,\ 3,\ 0.003325\alpha - 5.06\right)$$

$$\overset{(2)}{=} (4-4)^4 + \left(3-3\right)^2 + 4\left([-5.06 + 0.003525\alpha] + 5\right)^4$$

$$= 4\left(0.003525\alpha - 0.06\right)^4$$

Then $\quad g_2'(\alpha) = (16)(0.003525)(0.003525\alpha - 0.06)^3$

$g_2'(\alpha) = 0$ when $0.003525\alpha - 0.06 = 0$

We can solve directly for $g_2'(\alpha) = 0$ and obtain $\alpha_2 = \alpha = 16.29$.

$$\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2) = \langle 4, 3, -5.06 \rangle - 16.29 \langle 0, 0, -0.003525 \rangle$$

$$= \langle 4, 3, -5.002 \rangle$$

Because $\mathbf{x}_3$ and $\mathbf{x}_2$ are close, we stop.

$\vec{x}_2 = \langle 3, 4, -5.060 \rangle$

$\vec{x}_3 = \langle 3, 4, -5.002 \rangle$

Note: It is easy to see from the formula of f that the exact minimizer is $x^\star = \langle 4, 3, -5 \rangle$.

Exercise: Perform 2 iterations in the steepest descent method to find the minimizer of $f(x, y) = x + \frac{1}{2}x^2 + \frac{1}{2}y + y^2 + 3$ starting with $\mathbf{x}_0 = 0$. ← vector $(0,0)$

(1) $\vec{x}_0 = (x_0, y_0) = (0,0)$

Sol. (To save space, we write vectors in row form as in Calculus 2.)

$$\nabla f(\vec{x}) = \nabla f(x,y) = \langle 1 + x, \tfrac{1}{2} + 2y \rangle \qquad (2)$$

$$\vec{x} - \alpha \nabla f(\vec{x}) = (x,y) - \alpha \nabla f(x,y)$$

$$g(\alpha) = f((x,y) - \alpha \nabla f(x,y))$$

$\underline{k=0}$: $\nabla f(\vec{x}_0) = \nabla f(0,0) = \langle 1, \frac{1}{2} \rangle$ — (a)

$$\vec{x}_0 - \alpha \nabla f(\vec{x}_0) = (0,0) - \alpha(1, \tfrac{1}{2}) = \langle -\alpha, -\tfrac{\alpha}{2} \rangle$$

$$g_0(\alpha) = f(-\alpha, -\tfrac{\alpha}{2}) \underset{(1)}{=} (-\alpha) + \tfrac{1}{2}(-\alpha)^2 + \tfrac{1}{2}(-\tfrac{\alpha}{2}) + (-\tfrac{\alpha}{2})^2 + 3$$

$$= \tfrac{3}{4}\alpha^2 - \tfrac{5}{4}\alpha + 3$$

Find minimizer of g_0:

$g_0'(\alpha) = 0$ when $\frac{6}{4}\alpha - \frac{5}{4} = 0 \implies \alpha_0 = \alpha = \frac{5}{6}$

So $\vec{x}_1 = \vec{x}_0 - \alpha_0 \nabla f(\vec{x}_0) = (0,0) - \frac{5}{6}(1, \frac{1}{2}) = \underline{\underline{(-\frac{5}{6}, -\frac{5}{12})}}$ (a)

$\underline{\underline{k=1}}$: Start with $\vec{x}_1 = (x_1, y_1) = \left(-\frac{5}{6}, -\frac{5}{12}\right)$

$$\nabla f(\vec{x}_1) = \nabla f\left(-\frac{5}{6}, -\frac{5}{12}\right) \overset{(2)}{=} \left(1 - \frac{5}{6}, \frac{1}{2} + 2\left(-\frac{5}{12}\right)\right) = \left(\frac{1}{6}, -\frac{1}{3}\right)$$

$$\vec{x}_1 - \alpha \nabla f(\vec{x}_1) = \left(-\frac{5}{6}, -\frac{5}{12}\right) - \alpha\left(\frac{1}{6}, -\frac{1}{3}\right) = \left(-\frac{5}{6} - \frac{\alpha}{6}, -\frac{5}{12} + \frac{\alpha}{3}\right)$$

$$g_1(\alpha) = f\left(\vec{x}_1 - \alpha \nabla f(\vec{x}_1)\right) = f\left(\frac{-5-\alpha}{6}, \frac{-5+4\alpha}{12}\right)$$

$$\overset{(1)}{=} \left[\frac{-5-\alpha}{6}\right] + \frac{1}{2}\left[\frac{-5-\alpha}{6}\right]^2 + \frac{1}{2}\left[\frac{-5+4\alpha}{12}\right] + \left[\frac{-5+4\alpha}{12}\right]^2 + 3$$

$g_1'(\alpha) = 0$ when

$$-\frac{1}{6} + \frac{1}{\cancel{2}} \cdot \cancel{2}\left[\frac{-5-\alpha}{6}\right]\left[-\frac{1}{6}\right] + \frac{1}{2}\left[\frac{4}{12}\right] + 2 \cdot \left[\frac{-5+4\alpha}{12}\right] \cdot \left[\frac{4}{12}\right] = 0$$

$$\frac{[-\cancel{6} + 5 + \cancel{6} - 10] + [\alpha + 8\alpha]}{36} = 0$$

$$-5+9\alpha=0 \Rightarrow \alpha_1=\alpha=\frac{5}{9}$$

So $\vec{x}_2=\vec{x}_1-\alpha_1\nabla f(\vec{x}_1)=\left(-\frac{5}{6},-\frac{5}{12}\right)-\frac{5}{9}\left(\frac{1}{6},-\frac{1}{3}\right)$

$$=\left(\frac{-90-10}{108},\frac{-45+20}{108}\right)$$

$$=\left(\frac{-100}{108},\frac{-25}{108}\right)\approx(-0.926,-0.232)$$

<u>Remark:</u> From (2) we obtain:

$\nabla f(x_1,x_2)=0$ when $\vec{x}=\vec{x}^*=\left(-1,-\frac{1}{4}\right)$

fairly close